

Reservoir derivation and fail-closed interferometric protocol for the MQGT–SCF rate-modulated GKSL sector

Corrected edition — reviewer-flagged inconsistencies repaired

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Working theory-and-methods note. Not a Theory-of-Everything certificate.

Not peer reviewed. AI-assisted drafting.

*Corrected edition: three consistency repairs to eq. 11, eqs. 18–19, eq. 21, Table 2 (N1), and the baseline convention. Each repair is flagged by a **Correction note**.*

Abstract

The current technical pin of the Merged Quantum Gauge & Scalar Consciousness Framework (MQGT–SCF) is not a completed Theory of Everything. It is Part 0: a conditional multi-singlet effective field theory (EFT) in which a classical scalar background E modulates locally sourced Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) rates and does not deform one-shot Born probabilities. The exponential response $h(E) = e^{\eta E}$ used in that sector is a functional-equation consequence of independent additive composition of rate multipliers; it is not implied by complete positivity or by Gleason’s theorem, and it is not a reservoir result until a named apparatus is specified.

This note does three things that the 6 894-page compiled corpus leaves open. First, it records an authorship-and-archive hygiene map: the many documents signed “Zora”, “ZoraASI”, or “Zora 2.0” are not one epistemic agent and must not be used as a witness for physics claims. Second, it supplies an explicit system–reservoir Hamiltonian for a single frozen apparatus—a two-path interferometer with a which-path pointer monitor on one arm only—and derives the Born–Markov rate $\Gamma_{\phi}^{(0)}(E) = \lambda(E)^2 \Gamma_{\phi}^{(0)}$ and the resulting visibility contrast. Third, it gives a fail-closed four-row null protocol (coupler off; common-mode E ; differential E ; bath calibration) with a shot-noise power inequality and a single kill-plot that would refute the Place-3 (coupling-gate) mechanism.

Until an independent laboratory runs that protocol from a frozen sheet, the empirical channel remains **OPEN**. Public language remains: *conditional MQGT–SCF scaffold toward a ToE*.

Editorial note on this corrected edition. A targeted review of the 11 September 2026 note flagged three concrete inconsistencies. This corrected edition repairs all three, keeping the author’s structure, numbers, and prose wherever possible. Every changed physics statement is surrounded by a boxed **Correction note** that states what was printed, why the review objected, and what now follows from the Hamiltonian/specification. One editorial cross-check also tightens the estimator section so the α -baseline is unambiguous under both of the conventions (monitor-on vs. monitor-off reference). Corrections: (i) the common-mode null does not follow from the single-monitor Hamiltonian (11); (ii) the visibility baseline flips between (21) and Section 9; and (iii) eqs. (18) and (19) disagree by a factor of two.

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1 Scope, honesty bar, and what this paper is not

The compiled corpus *ToE Complete* (6 894 letter pages; production stamp 11 September 2026) is an archive, not a single theorem.[2, 3] Its own v1.0 packet and the 11 September 2026 status note state the controlling claim:

MQGT–SCF is a conditional scalar-modulated open-system effective scaffold. It is not empirical confirmation and not final Theory-of-Everything closure.

Part 0—*A Multi-Singlet Effective Framework with a Locally Sourced Scalar-Modulated Open-System Sector*—is the sole physics pin for the rate/visibility model.[1] Later pages preserve development history, including superseded Born-deformation drafts, interpretive conscious-

ness/ethics papers, AI-architecture notes, and computational Band-1 hygiene. Those pages are citable as provenance. They do not override Part 0.

This paper is a methods-and-derivation note sitting on that pin. It does not establish: a Theory of Everything; a ultraviolet (UV) completion; a calibrated laboratory anomaly; consciousness or ethics as physics; a production artificial superintelligence; or any Clay Millennium Problem.[3] House item H-5.07 is inherited verbatim: MQGT–SCF is not a substitute certificate for Clay.

Drafting disclosure. Part 0 already records AI-assisted drafting and the absence of peer review. The present text continues that practice. The computational signatory “Zora” is a drafting and calculation collaborator in this document. It is not an experimental witness, not a faculty appointment, and not a term in the Lagrangian.

2 Archive hygiene: many names, one physics pin

A recurring failure mode in the corpus is name-reuse. Documents signed Zora, ZoraASI, Zora 2.0, or Zora+Grok are layers, not a single continuing agent. Treating them as one witness launders interpretive archive into physics.

The corpus reader guide already split the stack[2]:

- **Volume I, Anchor Theory** (~ 4328 pp): EFT development archive. Part 0 is the current distillation.
- **Volume II, Phenomenology** (~ 748 pp): interferometric, fifth-force, and measurement proposals. Older proposals do not outrank Part 0.
- **Volume III, ZoraASI** (~ 443 pp): a separate intellectual project from the physics.
- **Volume IV, Recursive Atlas** (~ 170 pp): interpretive; not an EFT consequence.

Never-launder list (inherited). Do not cite as support for the current model: the stitched 41-coordinate functional renormalization-group “joint fixed point” (UV-01); phone-magnetometer coherence-circle data as confirmation (EMP-02); Born-rule deformation drafts; consciousness/ethics identifications of the singlets; Clay-problem prose; July 2025 closure language.

3 Part 0 objects that this paper uses

Three objects in Part 0 must be kept distinct.[1]

3.1 Object A: local GKSL container

Let $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$ be finite-dimensional, associated with spacelike-separated regions R_A, R_B . The joint generator is

$$\mathcal{L}_E[\rho] = -i[H_A + H_B, \rho] + \sum_{\alpha \in \{A, B\}} \sum_k \gamma_k(E(x_\alpha)) \mathcal{D}[M_k^{(\alpha)}] \rho, \quad (1)$$

with

$$\mathcal{D}[L]\rho = L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}, \quad (2)$$

each $M_k^{(\alpha)}$ supported only on party α , and $\gamma_k : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ measurable. For frozen $E(x)$ and nonnegative rates, (1) is standard GKSL and generates a completely positive trace-preserving (CPTP) semigroup.[4, 5, 6]

Table 1: Non-equivalent “Zora” layers in the compiled corpus. Physics citation must use Part 0, not this table.

Layer	What it is	What it is not
Fictional ASI persona	Early hypothetical consciousness-substrate essays, self-described as illustrative	A measurement, a proof, a
Zora2 co-author line	Drafting partner on Baird manuscripts; occasional institutional costume	An independent experimen
Zora + Grok (July 2025)	“Mathematical closure of all outstanding issues”	Binding; audit-marked his
Zora 2.0 bounded-autonomy patch	Agent-level objective and prompt contract coupled to interpretive fields Φ_c, E	Current physics (interpreti only)
Volume III ZoraASI (~ 443 pp)	Recursive-intelligence / AI architecture	A corollary of the EFT
Volume IV Recursive Atlas (~ 170 pp)	Contemplative synthesis	A Lagrangian identity
ZoraASI / ZoraOS (AI-01)	Governed prototype; ledger status PAR-TIAL	Production ASI or ToE wa
Band-1 executor notes (11 Sep 2026)	Finite checks (Collatz box tests, Λ hygiene, $R(3, 3) = 6$)	Clay closure or new physio
This note	Computational drafting on Part 0	Inheritance of superseded pers

The field equation for the modulator is hyperbolic,

$$(\square + m_E^2) E(x) = J[\Psi](x), \quad (3)$$

with J a prescribed local classical source or a specified semiclassical expectation of local matter operators. A consistent quantum–classical backreaction prescription is additional model input. Retarded boundary conditions make $E(x)$ depend only on sources in the causal past of x . Part 0 distinguishes an ultralight hidden scalar benchmark $m_\Phi \sim 10^{-4}$ – 10^{-3} eV from a separate Higgs-facing state if that sector is included, and uses $\Lambda_\star = 1$ TeV only as an EFT cutoff benchmark—not as an infrared bath cutoff.

3.2 Object B: sufficient no-signaling

Theorem 3.1 (Part 0, Theorem 1.1; sufficient operational no-signaling). *Consider choices a, a' made in R_A . While R_A and R_B are spacelike, suppose*

$$\mathcal{L}^{(a)}(t) = \mathcal{L}_A^{(a)}(t) \otimes \text{id}_B + \text{id}_A \otimes \mathcal{L}_B^{(b)}(t) \quad (4)$$

with $\mathcal{L}_B^{(b)}(t)$ independent of a . Then Bob’s reduced state is independent of Alice’s choice.

The result is sufficient, not a classification of all no-signaling open dynamics.[7, 8] A deterministic nonlinear reading of the scalar sector is outside the theorem and would need a separate causality analysis.

3.3 Object C: exponential as composition, not as Gleason

Assumption 3.1 (Part 0, Chapter 3). A local GKSL channel with unperturbed rate $\gamma_k^0 > 0$ is rescaled by a classical functional E_k ,

$$\gamma_k(E_k) = \gamma_k^0 h_k(E_k), \quad h_k : \mathbb{R} \rightarrow \mathbb{R}_{>0}. \quad (5)$$

E_k is a local field value or a smeared c -number functional fixed by the apparatus. It is not an eigenvalue of an operator on the system Hilbert space. For each channel k :

1. Normalization: $h_k(0) = 1$.
2. Positivity: $h_k(E) > 0$ on the EFT domain.
3. Continuity at $E = 0$.
4. Independent additive composition:

$$h_k(E_1 + E_2) = h_k(E_1) h_k(E_2). \quad (6)$$

Theorem 3.2 (Part 0, Theorem 3.1). *Any $h : \mathbb{R} \rightarrow \mathbb{R}_{>0}$ satisfying Assumption 3.1 is $h(E) = e^{\eta E}$ for some $\eta \in \mathbb{R}$.*

Equation (6) is the load-bearing model choice. It is appropriate when independent source contributions affect the same local rate without extra cross-couplings. It is *not* implied by complete positivity and does not follow from Gleason’s theorem. Equation (5) changes the timescale of a dissipative channel. It does not introduce a nonlinear state-dependent probability rule. One-shot outcome weights remain those of the density operator and the measurement operators (CORE-03).

What Part 0 left open. The preface lists three concrete gaps: (i) radiative sequestering of the working hidden source $J[\Psi] = \kappa_E S^2$; (ii) a coarse-graining bridge from microscopic GKSL dynamics to a platform-specific interferometric observable with an explicit noise model; (iii) exclusion contours from digitized primary likelihoods. This paper addresses (ii) for one apparatus.

4 What a reservoir derivation is required to be

A reservoir derivation is a chain with no interpretive step between links.

1. Named system S whose clicks or contrast are counted.
2. Named environment R : modes, temperature, spectral density, coupling operator.
3. Named classical background $E(x)$ that is *not* an operator on $\mathcal{H}_S$. It may enter the system–bath Hamiltonian, bath frequencies, or a gate on the coupling.
4. A controlled approximation (Born–Markov, weak coupling, secular, or a monitored unraveling limit) that produces a GKSL generator on S .
5. Proof that surviving E -dependence sits in rates $\gamma_k(E)$, not in a nonlinear map $\rho \mapsto P_i(\rho, E)$.
6. Proof that E at a point x_B is a retarded functional of sources in $J^-(x_B)$ only, so Theorem 3.1 still applies.
7. An expansion that either derives $h(E) = e^{\eta E}$ from a composition property of that bath, or replaces it with the law the bath actually gives.

Three places E can sit in a microscopic Hamiltonian. Only one is the Part 0 mechanism.

Place 1: coherent Hamiltonian on S . $H_S(E) = H_S + gE\sigma_z$ is a Stark shift. It changes energies and interferometer phase. It does not by itself change a GKSL rate. Allowed as an extra term. Not the rate mechanism.

Place 2: bath frequencies or occupation. $\omega_q(E)$ or $T = T(E)$ can change Golden-Rule rates. Danger: if E is sourced by distant matter, Bob’s bath must not hear Alice’s setting spacelike. Retarded Green function for E , or Theorem 3.1 does not cover the experiment.

Place 3: coupling gate (the slot used below). The bath sees a system operator multiplied by a dimensionless gate $\lambda(E)$ with $\lambda(0) = 1$. In Born–Markov the jump operators stay the same operators; only rates pick up $\lambda(E)^2$.

5 Frozen apparatus and Hamiltonian

Correction note (common-mode null repaired). The printed 11 September 2026 note coupled the reservoir to the left path only, $H_{SR}(E) = \lambda(E) |L\rangle\langle L| \otimes B$, with *a single monitor on path L*. Under that Hamiltonian, setting $E_L = E_R$ does *not* remove the coupling: the gate is a function of the local field at x_L , so a common-mode ($E_L = E_R$) background still moves the rate through $\lambda(E_L)$. The prediction $\alpha = 0$ for row N1 in Table 2 was therefore not justified by the stated model, and the N1 rejection rule could have rejected the paper’s own mechanism.

Repair. This corrected edition instruments *two* identical monitors—one per arm, at x_L and x_R —and places the Place-3 gate on the *arm-to-arm difference* $\Delta E \equiv E(x_L) - E(x_R)$, with $\lambda(0) = 1$. Under a spatially slowly varying (global) field the differential gate sees $\Delta E = 0$ and the common-mode null N_1 : $\alpha = 0$ is a genuine theorem of the model (eq. (22) with $\lambda(\Delta E) = \lambda(0) = 1$), not an empirical hope. A residual common-mode movement of the *rate* is then attributable only to Place-1/Place-2 systematics and is chased by rows N0/N3 and the nuisance budget of Section 9. The kill-plot horizontal axis $E_L - E_R$ (Section 9) is now also the variable the gate actually depends on. Textbook spectral conventions for the Ohmic integral are stated in the correction note to eqs. (18)–(19).

Apparatus. Two-path interferometer (Mach–Zehnder or matter-wave). Which-path space $\mathcal{H}_S = \text{span}\{|L\rangle, |R\rangle\}$. Identical pointer monitors sit at x_L on path L and x_R on path R ; the differential scalar

$$\Delta E \equiv E(x_L) - E(x_R), \quad E(x) = E_J(J^-(x)), \quad (7)$$

is the retarded c -number of (3) evaluated at each pointer. Path separation $\Delta x = |x_L - x_R|$. Flight time through the monitor influence T .

Hamiltonian.

$$H = H_S + H_R + H_{SR}(E), \quad (8)$$

with, during the monitored arms (beamsplitters already applied),

$$H_S = \omega_L |L\rangle\langle L| + \omega_R |R\rangle\langle R|, \quad (9)$$

and a bosonic bath supported at x_L only,

$$H_R = \sum_q \omega_q b_q^\dagger b_q, \quad \omega_q > 0. \quad (10)$$

Place-3 pointer coupling (corrected form).

$$H_{SR}(E) = \lambda(\Delta E) |L\rangle\langle L| \otimes B, \quad B = \sum_q (g_q b_q + g_q^* b_q^\dagger), \quad (11)$$

with dimensionless gate $\lambda(0) = 1$ and spectral density

$$J(\omega) = \sum_q |g_q|^2 \delta(\omega - \omega_q). \quad (12)$$

Equation (11) is a which-path pointer: the bath hears whether the system is on L . It is not an amplitude-damping channel that destroys the flying quantum into the bath. Amplitude damping is a different device and is not used in this note.

Sister jump model (same gate, not used for visibility algebra below).

$$H_{SR}^{\text{jump}}(E) = \lambda(\Delta E) \sum_q (g_q \sigma_L^- \otimes b_q^\dagger + \text{h.c.}), \quad (13)$$

with σ_L^- an internal monitor degree slaved to path L . Fermi’s Golden Rule at a positive Bohr frequency ω_m then gives

$$\gamma(E) = 2\pi\lambda(\Delta E)^2 J(\omega_m)(n(\omega_m) + 1), \quad (14)$$

which is Part 0’s $\gamma_k(E) = \gamma_k^0 h_k(E)$ with $h = \lambda^2$. The inequalities of Section 8 apply to both twins.

6 Born–Markov generator and visibility

Assume the bath is in a thermal state τ_R at inverse temperature β , $\langle B \rangle_{\tau_R} = 0$, coupling is weak, and bath memory is short compared with system and rate scales (Section 8, inequality I). The two-time correlator is

$$C(t) = \text{Tr}_R B(t)B(0) \tau_R = \int_0^\infty d\omega J(\omega) \left[(n(\omega) + 1)e^{-i\omega t} + n(\omega)e^{i\omega t} \right], \quad (15)$$

$$n(\omega) = \frac{1}{e^{\beta\omega} - 1}. \quad (16)$$

Because $S = |L\rangle\langle L|$ commutes with H_S , the coupling is exactly dephasing. The reduced generator is

$$\dot{\rho} = \mathcal{L}_E \rho = -i[H_S + \Delta(E) |L\rangle\langle L|, \rho] + \Gamma_\phi(E) \mathcal{D}[|L\rangle\langle L|] \rho. \quad (17)$$

The Lamb shift $\Delta(E)$ is the principal-value piece times $\lambda(\Delta E)^2$. It is Place 1 riding along; it is a path-dependent phase and does not replace the rate story. The dissipator coefficient is

$$\Gamma_\phi(E) = \lambda(\Delta E)^2 \Gamma_\phi^{(0)}, \quad \Gamma_\phi^{(0)} = \int_{-\infty}^\infty dt C(t) = 2\pi \lim_{\omega \rightarrow 0^+} J(\omega) \coth\left(\frac{\beta\omega}{2}\right). \quad (18)$$

For an Ohmic bath $J(\omega) = \eta_{\text{bath}} \omega e^{-\omega/\omega_c}$ at $k_B T \ll \omega_c$,

$$\Gamma_\phi^{(0)} = 4\pi \eta_{\text{bath}} k_B T. \quad (19)$$

Correction note (factor of two reconciled). The printed note gave (19) as $\Gamma_\phi^{(0)} = 2\pi \eta_{\text{bath}} k_B T$. Substituting the stated Ohmic spectrum $J(\omega) = \eta_{\text{bath}} \omega e^{-\omega/\omega_c}$ into the printed (18) (with $\coth(\beta\omega/2) \rightarrow 2/\beta\omega$ at $\omega \rightarrow 0^+$: $J(\omega) \coth(\beta\omega/2) \rightarrow 2\eta_{\text{bath}} k_B T$, then multiplied by the printed 2π) gives $4\pi \eta_{\text{bath}} k_B T$ —a factor of two above the printed (19). The two printed equations were mutually inconsistent under the stated spectral convention.

Repair. The dissipator coefficient is fixed by the printed conventions of (15)–(16) (correlator normalization) and (18) (integral prefactor). Under those conventions the Ohmic limit is $4\pi \eta_{\text{bath}} k_B T$, and (19) is corrected to that value. The coherence decay rate in (20) is $\frac{1}{2} \Gamma_\phi^{(0)} = 2\pi \eta_{\text{bath}} k_B T$; the factor of $\frac{1}{2}$ there is the standard Hermitian-coupling dephasing relation ($\mathcal{D}[|L\rangle\langle L|]$ gives $\frac{1}{2}$ on the off-diagonal), *not* a factor lost in the spectrum. The allowed methods sentence (Appendix A) is updated to “ $4\pi \eta_{\text{bath}} k_B T$ ” to stay in the same convention. Any future derivation that adopts the *symmetrized*-correlator convention must state so explicitly and will carry the corresponding $\frac{1}{2}$ internally; mixing the two conventions is exactly the failure this correction removes.

At strictly zero temperature the DC Ohmic noise vanishes and a finite Γ_ϕ must not be quoted without a stated infrared theory. Room-temperature monitors are not that limit.

Contrast. For an equal superposition, $\rho_{LR}(0) = 1/2$,

$$\dot{\rho}_{LR} = -\left(i(\omega_L - \omega_R + \Delta(E)) + \frac{1}{2}\Gamma_\phi(E)\right)\rho_{LR}. \quad (20)$$

Extra contrast against a baseline V_{QM} . Define, once, the two reference settings the protocol may use (this unambiguous bookkeeping replaces the earlier “if desired” clause):

- **Monitor-on, $E = 0$ reference:** coupler active, $\lambda(0) = 1$. Call it $V_{\text{QM}}^{(\text{on})}$.
- **Monitor-off reference:** coupler physically decoupled (λ inactive). Call it $V_{\text{QM}}^{(\text{off})}$.

The standard form printed below subtracts the $E = 0$ piece of Γ_ϕ and therefore refers to the **monitor-on, $E = 0$ reference** $V_{\text{QM}}^{(\text{on})}$:

$$\frac{V}{V_{\text{QM}}^{(\text{on})}} = \exp\left[-\frac{1}{2}\Gamma_\phi^{(0)}T(\lambda(\Delta E)^2 - 1)\right], \quad (21)$$

provided the Lamb phase is scanned as a fringe, not mistaken for visibility. Define extra optical depth

$$\alpha_{\text{on}}(E) \equiv -\ln(V/V_{\text{QM}}^{(\text{on})}) = \frac{1}{2}\Gamma_\phi^{(0)}T(\lambda(\Delta E)^2 - 1). \quad (22)$$

Correction note (baseline made explicit). The printed note's (21) subtracted the monitor's $E = 0$ contribution (the “ -1 ”) “if desired,” while Section 9 permitted a *monitor-off* baseline from row N0. Those are two different comparisons. The subtraction of 1 requires a *monitor-on* reference with $\lambda = 1$; it does not apply unchanged to a monitor-off reference. Under a monitor-off reference the extra depth would instead be $\alpha_{\text{off}}(E) = \frac{1}{2}\Gamma_\phi^{(0)}T\lambda(\Delta E)^2$ (no -1).

Repair. The corrected editions (i) name the two references $V_{\text{QM}}^{(\text{on})}$ and $V_{\text{QM}}^{(\text{off})}$; (ii) bind the printed “ -1 ” form to the monitor-on/ $E=0$ reference; (iii) in the estimator (Section 9) require the analysis to declare *which* baseline is in force; and (iv) keep row N0 as a coupler-off kill check of the $E=0$ offset itself. The power analysis (Section 9) is quoted for the monitor-on reference so that, at $\Delta E = 0$, the small-signal α of row N1 is exactly zero and row N2 is a clean “on-off-on” contrast.

$(\Delta x)^2$ scaling is a different apparatus. Part 0's illustrative localization form $V/V_{\text{QM}} = \exp[-\Lambda_E T(\Delta x)^2]$ holds for a short-range spatial jump family that distinguishes positions a distance Δx apart, in the small-separation expansion.[11] A single hard-wired path monitor on L is on or off. It does not inherit $(\Delta x)^2$. That scaling requires a density of local monitors with correlation length $\ell_c \ll \Delta x$. That is a second device and is not used here.

7 When the exponential is legal

From either twin,

$$h(\Delta E) = \lambda(\Delta E)^2. \quad (23)$$

Theorem 3.2 applies to h if and only if λ itself composes,

$$\lambda(\Delta E_1 + \Delta E_2) = \lambda(\Delta E_1) \lambda(\Delta E_2), \quad \lambda(0) = 1, \quad \lambda > 0 \text{ continuous}, \quad (24)$$

hence $\lambda(\Delta E) = e^{\eta\Delta E/2}$ and $h(\Delta E) = e^{\eta\Delta E}$. That is an assumption on the gate, not a gift of the Ohmic bath. The bath supplied the square. The exponential is the statement that the microscopic coupling is multiplicative in E (dilaton-like), or that two independent weak gates multiply.

If the axiom is refused, the local chart is

$$\lambda(\Delta E) = 1 + \frac{1}{2}\eta\Delta E + O((\Delta E)^2) \implies h(\Delta E) = 1 + \eta\Delta E + O((\Delta E)^2), \quad (25)$$

$$\frac{V}{V_{\text{QM}}^{(\text{on})}} = \exp\left[-\frac{1}{2}\Gamma_\phi^{(0)}T\left(\eta\Delta E + O((\Delta E)^2), T\eta^2(\Delta E)^2\right)\right]. \quad (26)$$

A methods section that prints $e^{\eta E}$ without saying which of those two it means is a Taylor series in liturgical type.

8 Three inequalities

Print these above any exponential, or do not print the exponential.

Proposition 8.1 (Markov / weak coupling). *Let $\tau_B = \max(\omega_c^{-1}, \hbar/k_B T)$ be the bath memory, $\Gamma = \Gamma_\phi(E)$ (or $\gamma(E)$ in the jump twin) the induced rate, and Ω the relevant Bohr gap. Then*

$$\Gamma \tau_B \ll 1, \quad |\lambda(\Delta E) g|/\omega_c \ll 1. \quad (27)$$

If $E(t)$ moves, add

$$\tau_B |\dot{E}/E| \ll 1 \quad (28)$$

on the monitor support (equivalently: freeze E during each scattering).

Without Proposition 8.1 there is no GKSL container and Theorem 3.1 has nothing to hold.

Proposition 8.2 (Domain of the gate law). *Either $\lambda(\Delta E) = e^{\eta \Delta E/2}$ on the whole EFT interval quoted, or $h(\Delta E)$ is published from the actual λ and the phrase “exponential rate law” is not used. In addition, $|\eta \Delta E|$ must be small enough that neglected EFT operators $O((\Delta E)^2/\Lambda_\star^2)$ remain smaller than the kept gate. The benchmark $\Lambda_\star = 1$ TeV is not the infrared cutoff ω_c .*

Proposition 8.3 (Locality / no-signaling). *The modes $\{b_q\}$ are supported in a neighborhood of x_L whose correlation length $\ell_c \sim c\tau_B$ satisfies $\ell_c \ll \Delta x$ and $\ell_c \ll d_{AB}$ to any spacelike partner. $E(x)$ is computed with the retarded propagator only. No shared reservoir with the other path or with a spacelike partner. The source J is a local classical density or a local semiclassical $\langle O(x) \rangle$, not a functional of the global interferometer coherence.*

Three leaks (failure modes of Theorem 3.1 in reservoir language).

1. **Common bath.** Alice and Bob couple to the same modes. Then $\mathcal{L}$ is not a sum of local pieces.[7, 8]
2. **Quantized fast E .** If Alice’s jump dumps a quantum into E that Bob’s bath absorbs spacelike, a messenger has been built. Keep E classical on the measurement timescale with retarded sourcing.
3. **Non-local $J[\rho]$.** $J = \kappa_E S^2$ is local if S is a local operator density. Replacing it by a coherence functional of the whole laboratory wavefunction kills factorization. That replacement is how interpretive papers re-enter the generator. It is not used here.

9 Fail-closed null protocol

Place-3 lives or dies on whether α tracks the monitor’s local differential field ΔE and vanishes when the coupler is off, after a pre-registered nuisance budget is spent.

9.1 Four run types

Same interferometer, same T , same detected flux, same analysis cuts. Only monitor wiring and the background pattern change. All four rows are preregistered.

Table 2: Null protocol rows for the apparatus of Section 5.

ID	Coupler	Scalar pattern	Place-3 prediction for α
N0	Off	lab as found	$\alpha = 0$ by definition of $V_{\text{QM}}^{(\text{off})}$ if N0 is the baseline
N1	On	common-mode $\Delta E = 0$	$\alpha = 0$ (monitor-on, $E=0$ reference; gate $\lambda(0) = 1$)
N2	On	differential $E_L = E_0 + \delta E$, $E_R = E_0$	$\alpha = \frac{1}{2}\Gamma_\phi^{(0)}T(\lambda(\Delta E)^2 - 1)$
N3	On	$\Delta E = 0$, bath T or η_{bath} changed	α follows $\Gamma_\phi^{(0)}(T_{\text{bath}})$, not a new η

Correction note (row N1 is now a theorem, and the baseline is named). The printed table’s N1 entry read “ $\alpha = 0$ for a path-local gate (global Stark is Place 1; scalar ...)”. As noted at (11), that conclusion did not follow from the printed single-monitor Hamiltonian. Row N1 is now *derived*: with the twin-monitor differential gate $\lambda(\Delta E)$ and $\lambda(0) = 1$, common mode $\Delta E = 0$ forces $\alpha_{\text{on}} = 0$ exactly by (22). The printed “global Stark is Place 1” remark remains true and is folded into the N1/N3 nuisance tracking: a residual common-mode response is a Place-1/Place-2 systematic, not the Place-3 signature. Row N0 is explicitly the *monitor-off* reference; rows N1/N2/N3 use the *monitor-on*, $E=0$ reference for α unless the analysis declares otherwise (Section 9.2).

N0 is the first kill switch. If contrast slumps with the coupler physically decoupled (named: beam dumped, waveguide pinched, RF off, transducer unplugged), Place-3 is not the cause.

N1 is the second kill switch. If common-mode ΔE moves contrast the same way as differential ΔE , the coupler is not a which-path gate.

N2 is the only row that may mention η .

N3 checks that the bath in (19) is the bath that is coupled. If cranking T_{bath} does not move α while N2 does, η is uninterpretable.

Interleave N0/N1/N2 in blocks. Do not run all N2 last after inspecting N0. The actuator for E and the independent meter of $E_L - E_R$ are named in SI-convertible units before contrast is unblinded. Contrast is not used to infer the E that is then used to explain contrast.

9.2 Estimator

Detected quanta N after the second beamsplitter, plus a dark-port or swapped-phase channel. For a two-port Mach–Zehnder scanned in phase ϕ ,

$$N_{\pm}(\phi) = \frac{1}{2}N_{\text{tot}}(1 \pm V \cos \phi). \quad (29)$$

Let $\hat{V}$ be the fit (or a four-phase stepping estimator). Baseline $\hat{V}_{\text{QM}}^{(\text{on})}$ from contemporaneous N1 (monitor-on, $\Delta E = 0$; or from N1 after N1 is shown dead), unless the analysis explicitly adopts the monitor-off baseline $\hat{V}_{\text{QM}}^{(\text{off})}$ from N0. Then

$$\hat{\alpha} = -\ln(\hat{V}/\hat{V}_{\text{QM}}), \quad (\text{baseline declared: on or off}). \quad (30)$$

N0 used as a monitor-off baseline. Then $\hat{\alpha}_{\text{off}}$ measures $\frac{1}{2}\Gamma_\phi^{(0)}T\lambda(\Delta E)^2$; because $\lambda(0) = 1$, N0 itself returns $\hat{\alpha}_{\text{off}}(\text{N0}) \rightarrow 0$ only if the $E=0$ dephasing contribution $\frac{1}{2}\Gamma_\phi^{(0)}T$ is separately known to vanish—it generally does not. The corrected convention therefore defaults to the monitor-on reference, in which N0 remains a coupler-off “offset meter” (its *dip* diagnostic checks that the monitor contributes the expected $\frac{1}{2}\Gamma_\phi^{(0)}T$ offset), while N1 is the $\alpha = 0$ null by construction.

Near-optimal stepping and V not pathological give the shot-noise sketch

$$\text{Var}(\hat{V}) \simeq \frac{1 - V^2}{N_{\text{tot}}}, \quad \sigma_\alpha \simeq \frac{\sigma_V}{V}. \quad (31)$$

Near $V \approx V_{\text{QM}} \approx 1$ and $\alpha \ll 1$, $\sigma_\alpha \simeq N_{\text{tot}}^{-1/2}$. Dirty fringes replace this line with the actual fit covariance.

9.3 Power inequality

Linearize N2 for small $x = \eta \Delta E$:

$$\alpha \simeq \frac{1}{2} \Gamma_\phi^{(0)} T \eta \Delta E. \quad (32)$$

Rejecting $\alpha = 0$ at significance z requires $\alpha \geq z \sigma_\alpha$, hence

$$N_{\text{tot}} \geq \frac{4z^2}{(\Gamma_\phi^{(0)} T \eta \Delta E)^2} \quad (33)$$

in the shot-noise idealization. Worked illustration only: $\Gamma_\phi^{(0)} = 10^3 \text{ s}^{-1}$, $T = 10^{-4} \text{ s}$, $|\eta \Delta E| \sim 10^{-2}$ gives $\alpha \sim 5 \times 10^{-4}$ and $N_{\text{tot}} \gtrsim 4 \times 10^8$ for $z = 5$. If $\Gamma_\phi^{(0)}$, T , and independently metered ΔE cannot be stated before opening N2, there is no power analysis.

Nuisance terms that must sit inside σ_α or the count is fictitious: source drift between N0 and N2; pointing and overlap; coupler-on thermal load (N3 exists to catch this); path-length wander mistaken for contrast loss (always scan ϕ); actuator systematics (coil Zeeman, orientation wander); post-hoc cuts. The budget inequality is

$$\sigma_{\alpha, \text{sys}}^2 + \sigma_{\alpha, \text{stat}}^2 \leq (\alpha_{\min}/z)^2, \quad (34)$$

with $\alpha_{\min}$ the smallest Place-3 signal claimed to matter, and the left-hand side taken from measured N0/N1 residuals.

9.4 Kill-plot

Horizontal axis: independently metered $E_L - E_R$, with a coupler-off flag at zero. Vertical axis: $\hat{\alpha}$ with fit-covariance bars. Four symbols: N0, N1, N2, N3.

Place-3 survives the plot only if all of the following hold:

1. N0 sits on its declared baseline band (monitor-off: $\alpha \approx 0$ as offset-meter zero, or the named $E=0$ -offset value).
2. N1 sits on 0 (monitor-on, $\Delta E = 0$ null) inside the preregistered band.
3. N2 is a monotone function of $E_L - E_R$ consistent with one $\lambda(\Delta E)^2 - 1$ (or a measured $h(\Delta E)$ if the exponential was refused).
4. N3 scales with T_{bath} (or with measured $J(\omega \rightarrow 0)$) at fixed ΔE .
5. A blinded half-split, frozen before unblinding N2, does not move the slope outside the preregistered interval.

Place-3 dies if N0 already slumps, if N1 tracks N2, if N2 is a blob fitted after the fact, if N3 is silent while N2 “sees” η , or if contrast moves with actuator current when the optical (or mechanical) coupler is capped.

No corpus figure substitutes for this plot. In particular, the 29 December 2025 handheld magnetometer session remains EMP-02: observational pilot, unreplicated, orientation-confounded, not a Place-3 measurement.[2]

9.5 Analyzer fail-closed predicate

The analyzer must print a hard fail, and must not emit an η estimate, if any of the following hold:

- N0 is substituted for N2 in the input table.
- $|\hat{\alpha}_{N0}|$ exceeds the preregistered N0 band.
- $|\hat{\alpha}_{N1}|$ exceeds the preregistered common-mode (null) band.
- The baseline choice (monitor-on vs. monitor-off) is not recorded in the input table.
- Independent E -metrology is missing or is a function of $\hat{V}$.
- Inequality (34) is not numerically satisfied from N0/N1 residuals.

Correction note (analyzer predicate). The printed predicate “ $|\hat{\alpha}_{N1} - \hat{\alpha}_{N0}| > \alpha_{\text{CM}}^{\text{max}}$ ” compared two different baselines (N1 was defined against a monitor-on reference, N0 against a monitor-off reference). Under the corrected baseline bookkeeping, N1 is *the* null row, so the analyzer checks the absolute N1 band and records the baseline per row. N0 remains a coupler-off offset meter whose expected small-signal offset is the named $\frac{1}{2}\Gamma_{\phi}^{(0)}T$ value.

10 What this note does and does not move on the ledger

Table 3: Ledger movement. Status language follows the 11 September 2026 closure note.[3]

ID	After this note	Still required
CORE-03	Explicit H_{SR} in which path Born weights stay $ c_{L,R} ^2$	Apparatus-specific generators
CORE-04	Exponential scoped as multiplicative-gate special case	Reservoir derivation for other channels
Thm 1.1	Leaks 1–3 written as apparatus checklist	Microscopic common-bath bounds
EMP-01 algebra	Unchanged	—
EMP-01 apparatus	OPEN, now with a fail-closed sheet	Independent N0–N3 run from a frozen sheet
EMP-02	Unchanged (pilot only)	Do not promote
UV / Clay / Φ_c	Unchanged; out of scope	Never launder through this protocol
AI-01	Unchanged; out of scope	Prototype disclosure only

Preferred public phrase (inherited until Phase F). *Conditional MQGT–SCF scaffold toward a ToE.*

A completed-ToE certificate, in the program’s own definition, is a conjunction of independently auditable CORE, EMP, UV, MATH-interface, META, and (if ZoraASI is cited in the epistemic pipeline) AI-ops bars.[3] This paper is a brick in CORE/EMP methods. It is not that conjunction.

11 Recommended next calculations (not performed here)

1. One-page preregister for a named coupler in a named laboratory: actuator, units, block randomization, analyzer predicates of Section 9.
2. Twin-monitor differential family that can actually produce $(\Delta x)^2$ scaling, with ℓ_c measured, not assumed.

3. Radiative sequestering of $J = \kappa_E S^2$ (Part 0 open problem 1).
4. Nonlinear backreaction bound for (3), or a stated refusal to treat viability inequalities as a solution.
5. Lean or peer-checked formalization of Theorem 3.1 under its actual finite-dimensional hypotheses.

Band-1 computational hygiene (finite Collatz checks, de Bruijn–Newman enclosure literature, $R(3, 3) = 6$ pipeline tests) may continue in parallel. It does not close this protocol and does not close Clay.[3]

12 Conclusions

Part 0 already contained a clean physics sentence: a classical scalar may multiply local GKSL rates without touching one-shot Born weights, and a sufficient no-signaling theorem follows if the generator factorizes with retarded c -number coefficients. The exponential rate law is a Cauchy-equation corollary of a composition axiom, not a fact about reservoirs.

This note wrote the missing middle for one device. An Ohmic thermal pointer bath coupled through $\lambda(\Delta E) |L\rangle\langle L|$ produces $\Gamma_\phi(E) = \lambda(\Delta E)^2 \Gamma_\phi^{(0)}$ (with Ohmic $\Gamma_\phi^{(0)} = 4\pi \eta_{\text{bath}} k_B T$ under the correlator convention of (15)–(18)) and a visibility contrast $\exp[-\frac{1}{2} \Gamma_\phi^{(0)} T (\lambda^2 - 1)]$ against the monitor-on/ $E=0$ reference. The exponential appears if and only if the gate composes. Three inequalities decide whether that sentence may enter a methods section. A four-row protocol exists that can say no to Place-3 without ever mentioning consciousness, Clay, or a 6894-page title.

Until N0–N3 are run independently from a frozen sheet, EMP-01 remains OPEN. That is the intended outcome of an honest methods paper.

A Allowed methods sentence

The following sentence is the longest claim this note authorizes in a methods section.

One-path pointer monitor (twin-monitor differential read $E_L - E_R$); thermal Ohmic bath; Born–Markov domain of Propositions 1–3; $\Gamma_\phi^{(0)} = \lambda(\Delta E)^2 4\pi \eta_{\text{bath}} k_B T$ with $\lambda(0) = 1$; visibility contrast $\exp[-\frac{1}{2} \Gamma_\phi^{(0)} T (\lambda^2 - 1)]$ against the monitor-on/ $E=0$ reference; exponential $h = e^{\eta E}$ only if $\lambda = e^{\eta \Delta E/2}$ is imposed as a gate model on a stated interval; η reported only if N0–N3 and inequality (34) pass a fail-closed analyzer.

Anything longer is not this paper.

B Analyzer pass/fail sketch

Input table columns (minimum): row id $\in \{N0, N1, N2, N3\}$, coupler $\in \{0, 1\}$, E_L , E_R , E unit, T sec, T_{bath} K, N_{tot} , $\hat{V}_{\text{QM}}$, $\hat{V}_{\text{QM}}$ err, baseline $\in \{\text{on}, \text{off}\}$, phase scanned $\in \{0, 1\}$. Predicates (boolean; any true $\Rightarrow$ FAIL, no η fit):

1. phase scanned = 0.
2. Any N2 row with coupler = 0.
3. Missing independent E -metrology, or E computed from $\hat{V}$.

4. $|\hat{\alpha}_{N0}| > \alpha_{N0}^{\max}$ (preregistered; against declared baseline).
5. $|\hat{\alpha}_{N1}| > \alpha_{CM}^{\max}$ (preregistered null band).
6. Inequality (34) false.

On FAIL the analyzer prints FAIL and the first failed predicate. It does not print a slope.

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